(Please write your Enrolment No. immediately) Enrolment No. _____

MID-TERM EXAMINATION

B. TECH PROGRAMMES (UNDER THE AEGIS OF USCIT)

6th SEMESTER, MARCH,2025

Paper Code: FSD-318T **Subject: Advanced Java Programming**

Time: 1.5 Hrs. **Max. Marks :30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1 (a)	Explain the key features of Java that make it a platform-independent language.	2	1
(b)	Describe the working of an applet in Java. How is it different from a standalone application?	2	1
(c)	How is a server implemented in Java? Discuss with an example.	2	1
(d)	How do stateful and stateless session beans differ?	2	3
(e)	Describe the role of HttpServletRequest and HttpServletResponse in servlet programming.	2	2
2 (a)	Write a Java program to demonstrate multithreading using the Runnable interface. Explain its execution flow.	5	1
(b)	Explain socket programming in Java with an example. How does a client communicate with a server using sockets?	5	1
3 (a)	Write a Java Servlet program to demonstrate session tracking using cookies. Explain how cookies help in maintaining session states.	5	2
(b)	Compare and contrast Servlets and JSP. Discuss their advantages and disadvantages in web development.	5	2
4 (a)	Explain the concept of exception handling in Java with an example. How does the try-catch-finally block work?	5	1
(b)	Write a Java program to create a simple Java Bean with properties and methods. Explain the importance of Java Beans in component-based development.	5	3